```
clc;
clear;
close all;
```

1. INPUT PARAMETERS

```
Fs = 1000;  % Sampling frequency (Hz)
Fp = 100;   % Passband edge (Hz)
Fst = 150;  % Stopband edge (Hz)
Rp = 1;     % Passband ripple (dB)
Rs = 40;    % Stopband attenuation (dB)

% Normalized frequencies
Wp = Fp/(Fs/2);
Wst = Fst/(Fs/2);
```

2. CHEBYSHEV TYPE-I FORMULA

```
epsilon = sqrt(10^(Rp/10)-1);

N_formula = acosh(sqrt((10^(Rs/10)-1)/(10^(Rp/10)-1))) ...
    / acosh(Fst/Fp);

N_theory = ceil(N_formula);
```

3. BUTTERWORTH FILTER

```
[Nb,Wb] = buttord(Wp,Wst,Rp,Rs);
[bB,aB] = butter(Nb,Wb,'low');
```

4. CHEBYSHEV TYPE-I FILTER

```
[Nc,Wc] = cheb1ord(Wp,Wst,Rp,Rs);
[bC,aC] = cheby1(Nc,Rp,Wc,'low');
```

5. ELLIPTIC FILTER

```
[Ne,We] = ellipord(Wp,Wst,Rp,Rs);
[bE,aE] = ellip(Ne,Rp,Rs,We,'low');
```

6. DISPLAY INPUT PARAMETERS

```
fprintf('\n=================================================\n');
fprintf(' IIR FILTER COMPARATIVE ANALYSIS\n');
fprintf('=================================================\n');

fprintf('\nINPUT PARAMETERS\n');
fprintf('Sampling Frequency = %.2f Hz\n',Fs);
```

```
fprintf('Passband Edge Frequency = %.2f Hz\n',Fp);
fprintf('Stopband Edge Frequency = %.2f Hz\n',Fst);
fprintf('Passband Ripple = %.2f dB\n',Rp);
fprintf('Stopband Attenuation = %.2f dB\n',Rs);
```

```
==================================================
 IIR FILTER COMPARATIVE ANALYSIS
==================================================

INPUT PARAMETERS
Sampling Frequency = 1000.00 Hz
Passband Edge Frequency = 100.00 Hz
Stopband Edge Frequency = 150.00 Hz
Passband Ripple = 1.00 dB
Stopband Attenuation = 40.00 dB
```

7. CHEBYSHEV FORMULA RESULTS

```
fprintf('\nCHEBYSHEV TYPE-I FORMULA\n');
fprintf('Ripple factor epsilon = %.4f\n',epsilon);
fprintf('Calculated order = %.4f\n',N_formula);
fprintf('Theoretical minimum N = %d\n',N_theory);
```

```
CHEBYSHEV TYPE-I FORMULA
Ripple factor epsilon = 0.5088
Calculated order = 6.2071
Theoretical minimum N = 7
```

8. FILTER ORDER COMPARISON

```
fprintf('\n==================================================\n');
fprintf(' FILTER ORDER COMPARISON\n');
fprintf('==================================================\n');

fprintf('Butterworth Order = %d\n',Nb);
fprintf('Chebyshev Type-I Order = %d\n',Nc);
fprintf('Elliptic Order = %d\n',Ne);
```

```
==================================================
 FILTER ORDER COMPARISON
==================================================
Butterworth Order = 12
Chebyshev Type-I Order = 6
Elliptic Order = 4
```

9. COEFFICIENTS

```
fprintf('\n==================================================\n');
fprintf(' BUTTERWORTH COEFFICIENTS\n');
```

```
fprintf('=====================================================\n');

fprintf('Numerator b =\n');
disp(bB);

fprintf('Denominator a =\n');
disp(aB);

fprintf('\n=====================================================\n');
fprintf(' CHEBYSHEV TYPE-I COEFFICIENTS\n');
fprintf('=====================================================\n');

fprintf('Numerator b =\n');
disp(bC);

fprintf('Denominator a =\n');
disp(aC);

fprintf('\n=====================================================\n');
fprintf(' ELLIPTIC COEFFICIENTS\n');
fprintf('=====================================================\n');

fprintf('Numerator b =\n');
disp(bE);

fprintf('Denominator a =\n');
disp(aE);
```

```
=====================================================
 BUTTERWORTH COEFFICIENTS
=====================================================
Numerator b =
   1.0e-03 *

  Columns 1 through 7

    0.0002    0.0027    0.0146    0.0488    0.1098    0.1757    0.2050

  Columns 8 through 13

    0.1757    0.1098    0.0488    0.0146    0.0027    0.0002

Denominator a =
  Columns 1 through 7

    1.0000   -6.8844   22.4341  -45.5100   63.7725  -64.8501   48.9643

  Columns 8 through 13

  -27.6092   11.5218   -3.4662    0.7128   -0.0899    0.0053


=====================================================
```

```
 CHEBYSHEV TYPE-I COEFFICIENTS
=====================================================
Numerator b =
   1.0e-03 *

    0.0464    0.2782    0.6956    0.9274    0.6956    0.2782    0.0464

Denominator a =
    1.0000   -4.8694   10.3807  -12.3367    8.5969   -3.3268    0.5586


=====================================================
 ELLIPTIC COEFFICIENTS
=====================================================
Numerator b =
    0.0197   -0.0171    0.0333   -0.0171    0.0197

Denominator a =
    1.0000   -3.0330    3.8118   -2.2911    0.5554
```

10. FREQUENCY RESPONSES

```
N = 4096;

[HB,f] = freqz(bB,aB,N,Fs);
[HC,~] = freqz(bC,aC,N,Fs);
[HE,~] = freqz(bE,aE,N,Fs);

MB = 20*log10(abs(HB)+eps);
MC = 20*log10(abs(HC)+eps);
ME = 20*log10(abs(HE)+eps);
```

11. PASSBAND RIPPLE

```
P = f <= Fp;

Ripple_B = max(MB(P))-min(MB(P));
Ripple_C = max(MC(P))-min(MC(P));
Ripple_E = max(ME(P))-min(ME(P));
```

12. STOPBAND ATTENUATION

```
S = f >= Fst;

Stop_B = -max(MB(S));
Stop_C = -max(MC(S));
Stop_E = -max(ME(S));
```

13. PERCENTAGE DEVIATION

```
Dev_C_Ripple = abs(Ripple_C-Rp)/Rp*100;
Dev_B_Stop = abs(Stop_B-Rs)/Rs*100;
Dev_C_Stop = abs(Stop_C-Rs)/Rs*100;
Dev_E_Stop = abs(Stop_E-Rs)/Rs*100;
```

14. TRANSITION CHARACTERISTICS

```
TransitionBW = Fst-Fp;
```

15. DISPLAY PERFORMANCE

```
fprintf('\n=====================================================\n');
fprintf(' FILTER PERFORMANCE\n');
fprintf('=====================================================\n');

fprintf('\nPASSBAND RIPPLE\n');
fprintf('Butterworth = %.4f dB\n',Ripple_B);
fprintf('Chebyshev Type-I = %.4f dB\n',Ripple_C);
fprintf('Elliptic = %.4f dB\n',Ripple_E);

fprintf('\nSTOPBAND ATTENUATION\n');
fprintf('Butterworth = %.4f dB\n',Stop_B);
fprintf('Chebyshev Type-I = %.4f dB\n',Stop_C);
fprintf('Elliptic = %.4f dB\n',Stop_E);

fprintf('\nCHEBYSHEV RIPPLE DEVIATION\n');
fprintf('Specified Ripple = %.2f dB\n',Rp);
fprintf('Measured Ripple = %.4f dB\n',Ripple_C);
fprintf('Percentage Deviation = %.4f %%\n',Dev_C_Ripple);

fprintf('\nTRANSITION CHARACTERISTICS\n');
fprintf('Passband Edge = %.2f Hz\n',Fp);
fprintf('Stopband Edge = %.2f Hz\n',Fst);
fprintf('Transition Band = %.2f Hz\n',TransitionBW);
```

```
=====================================================
 FILTER PERFORMANCE
=====================================================

PASSBAND RIPPLE
Butterworth = 0.8033 dB
Chebyshev Type-I = 1.0000 dB
Elliptic = 1.0000 dB

STOPBAND ATTENUATION
Butterworth = 40.0198 dB
Chebyshev Type-I = 41.3365 dB
Elliptic = 40.0000 dB
```

```
CHEBYSHEV RIPPLE DEVIATION
Specified Ripple = 1.00 dB
Measured Ripple = 1.0000 dB
Percentage Deviation = 0.0000 %

TRANSITION CHARACTERISTICS
Passband Edge = 100.00 Hz
Stopband Edge = 150.00 Hz
Transition Band = 50.00 Hz
```

16. MAGNITUDE RESPONSE

```
figure;

plot(f,MB,'LineWidth',1.5);
hold on;
plot(f,MC,'LineWidth',1.5);
plot(f,ME,'LineWidth',1.5);

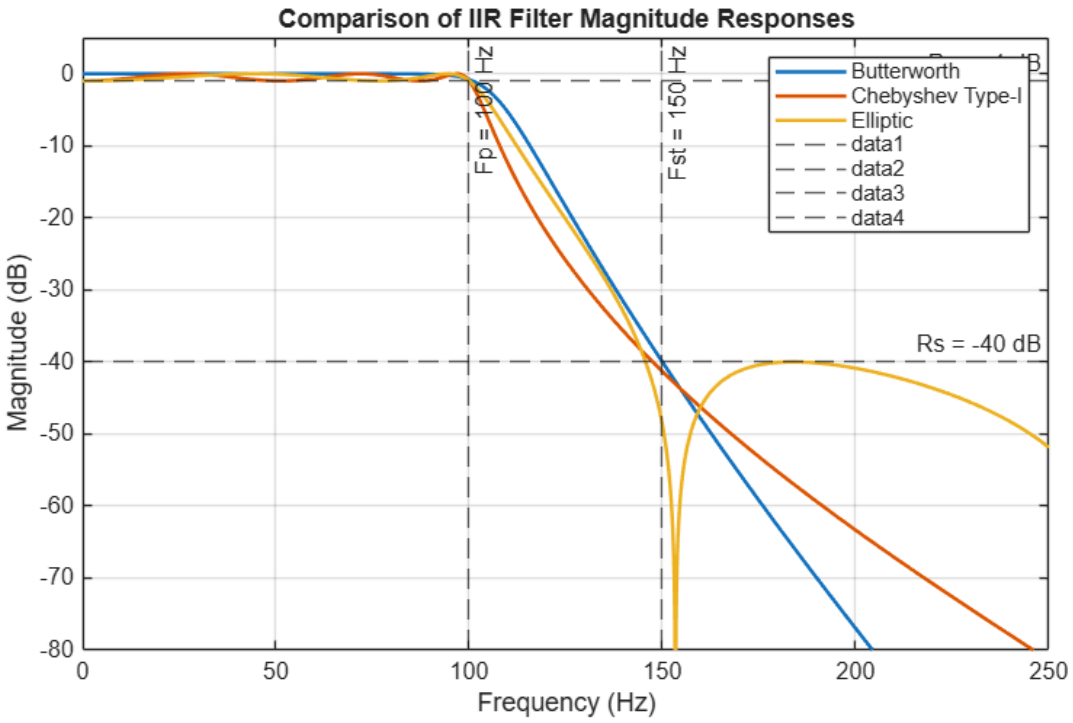
grid on;
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');
title('Comparison of IIR Filter Magnitude Responses');

legend('Butterworth','Chebyshev Type-I','Elliptic');

xlim([0 250]);
ylim([-80 5]);

xline(Fp,'--','Fp = 100 Hz');
xline(Fst,'--','Fst = 150 Hz');
yline(-Rp,'--','Rp = -1 dB');
yline(-Rs,'--','Rs = -40 dB');

hold off;
```



17. PHASE RESPONSE

```
Phase_B = unwrap(angle(HB))*180/pi;
Phase_C = unwrap(angle(HC))*180/pi;
Phase_E = unwrap(angle(HE))*180/pi;

figure;

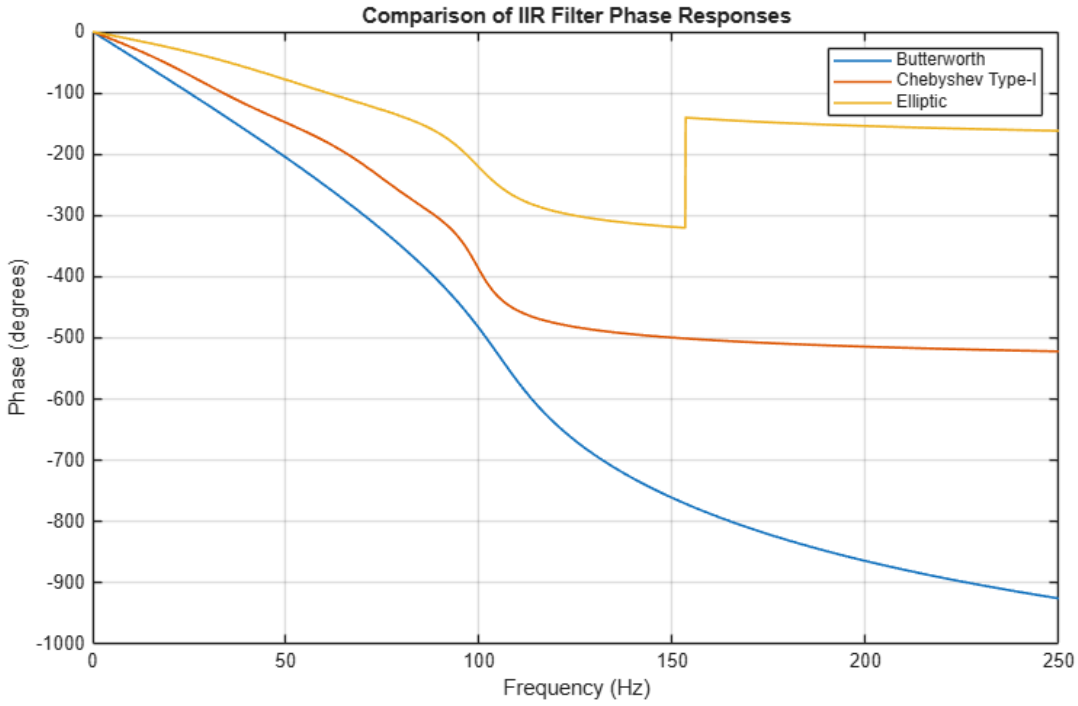
plot(f,Phase_B,'LineWidth',1.2);
hold on;
plot(f,Phase_C,'LineWidth',1.2);
plot(f,Phase_E,'LineWidth',1.2);

grid on;
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
title('Comparison of IIR Filter Phase Responses');

legend('Butterworth','Chebyshev Type-I','Elliptic');

xlim([0 250]);

hold off;
```



18. FILTER ORDER GRAPH

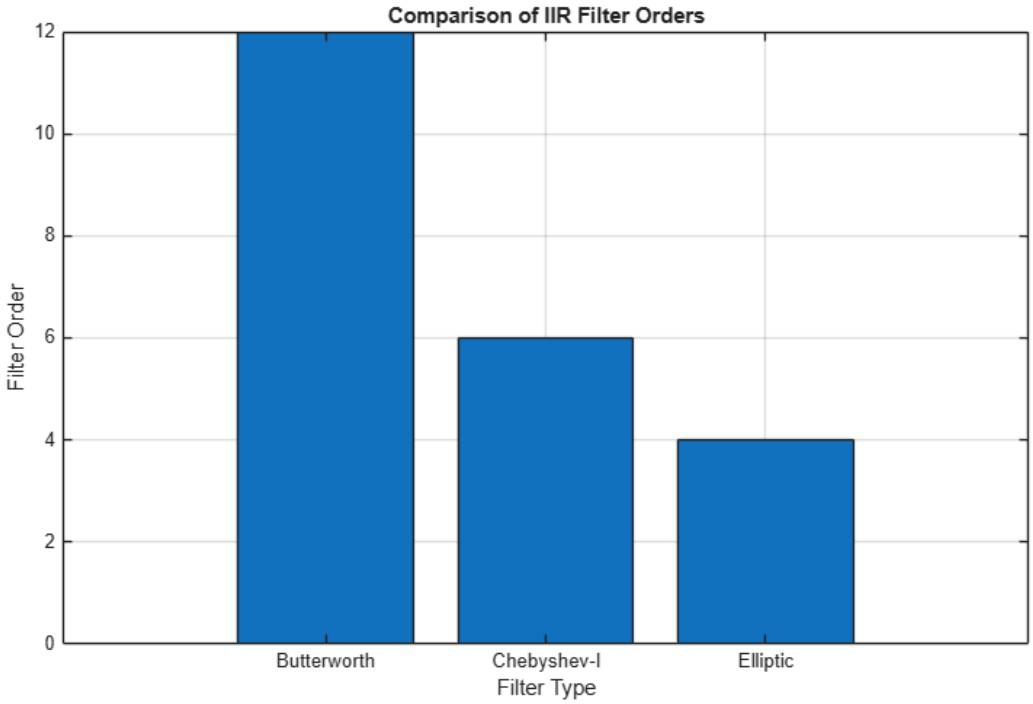
```
figure;

bar([Nb Nc Ne]);

set(gca,'XTick',1:3);
set(gca,'XTickLabel', ...
    {'Butterworth','Chebyshev-I','Elliptic'});

ylabel('Filter Order');
xlabel('Filter Type');
title('Comparison of IIR Filter Orders');

grid on;
```



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